



SEMINAR PAPER

Urban Land-Use Efficiency

A Stable Metric for SDG 11.3.1 and Its Economic Drivers

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Abstract

The official SDG 11.3.1 indicator, the ratio of land consumption rate to population growth rate (LCRPGR), is unstable: it divides by a growth rate that approaches zero and mixes an arithmetic numerator with a logarithmic denominator. We propose the Built-up per Capita Rate (BpCR), the log change in built-up area per person, which equals the difference of two log growth rates and is therefore well-defined for every country-period. Using a harmonised global panel of 193 UN member states (1985–2020) built from GHSL built-up and population grids under the Degree of Urbanisation, we (i) document where LCRPGR breaks down, (ii) show BpCR reproduces the policy signal without the instability, and (iii) estimate its economic drivers with two-way fixed-effects and dynamic-panel GMM models.

Keywords: SDG 11.3.1, Urban land-use efficiency, Built-up per capita rate (BpCR), LCRPGR, Dynamic panel GMM, GHSL / Degree of Urbanisation

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1. Introduction

The world is in the midst of the largest urban expansion in history. Every single year, the global built-up area grows by roughly the size of the Netherlands as housing, roads, and industry spread outward (Seto et al., 2012). Urbanisation itself is not the problem — cities must grow to house more people — but the policy question is *how much* land each new resident consumes. If land grows at the same pace as population, the city stays compact and infrastructure costs remain manageable. If land grows substantially faster than population, the city sprawls, and sprawl carries measurable costs: higher per-capita expenditure on roads, water, and transit; the systematic displacement of poorer residents to the periphery; permanent farmland loss; and locking cities into car-dependent, high-emission growth paths (Oueslati et al., 2015).

To track this, the United Nations adopted Sustainable Development Goal (SDG) target 11.3, which asks countries to “enhance inclusive and sustainable urbanisation” by keeping land consumption in line with population growth. Indicator 11.3.1 operationalises this target as the Ratio of Land Consumption Rate to Population Growth Rate (LCRPGR), a single metric intended to signal whether cities are densifying, balancing, or sprawling (UN-Habitat, 2025).

Yet the official indicator carries a structural flaw: it divides an arithmetic (linear) land growth rate by a logarithmic (exponential) population growth rate, creating a ratio that is dimensionally inconsistent and that explodes to $\pm\infty$ whenever population growth approaches zero — a common occurrence in shrinking or demographically stable cities (Nicolau et al., 2019). A city with identical 5% growth in both built-up area and population produces $\text{LCRPGR} = 1.13$, not 1.0, because $(g - 1) > \ln(g)$. The same 2% land expansion reads as 1.33, -20 , or 2000 depending only on how close the denominator is to zero. The UN metadata itself acknowledges that LCRPGR is “difficult to interpret” and proposes secondary indicators built from the same inputs, including built-up area per capita (UN-Habitat, 2025, §4.f).

The literature on urban land expansion has developed along two largely separate tracks. On the measurement side, Nicolau et al. (2019) document the instability of the official ratio and recommend a log-based formulation; Lu and Weng (2026) demonstrate that land-use efficiency estimates swing widely with dataset choice and argue for a log-based framework for reliable monitoring. On the driver side, Seto et al. (2011) show in a meta-analysis of 326 remote-sensing studies that urban land grew faster than population almost everywhere, with income as the dominant driver across most regions; Oueslati et al. (2015) find that rising income and population both expand the urban footprint in European cities, and that cities continue to spread even where population is flat.

We bridge these two strands. Following the log-based approach recommended by Nicolau et al. (2019) and Lu and Weng (2026), we construct the Built-up per Capita Rate (BpCR) — the annual log growth rate of built-up area per capita, defined as the difference of two logarithmic rates:

$$\text{BpCR} = \frac{1}{z} \ln \left(\frac{V_t/P_t}{V_{t-z}/P_{t-z}} \right) = \text{LCR}_{\log} - \text{PGR}_{\log}.$$

Unlike LCRPGR, BpCR places numerator and denominator on the same mathematical scale, is well-defined for every country-period including those with stable or shrinking populations, and retains an intuitive interpretation: positive values signal sprawl (per-capita land use rising), negative values signal densification (per-capita land use falling).

We then make the metric *dynamic*. Using a harmonised global panel of 193 UN member states spanning eight five-year periods (1985–2020) — built from GHSL built-up and population grids under the Degree of Urbanisation — we estimate a two-way fixed-effects model with a lagged dependent variable to capture path-dependence in per-capita land use, and we correct the resulting Nickell bias with Arellano-Bond and Blundell-Bond dynamic-panel GMM. This allows us to answer three questions: (i) does the choice of metric change the empirical conclusions? (ii) what economic and structural factors drive within-country changes in per-capita urban land use? and (iii) how persistent are these trajectories?

We find that the metric matters decisively. The official LCRPGR produces a within-country R^2 of approximately 0.003, with no variable reaching statistical significance. The same specification estimated with BpCR yields a within-country R^2 roughly twenty times higher, and clear, significant drivers emerge. Our GMM-corrected estimates reveal strong path-dependence ($\rho \approx 0.6$): roughly 60% of the previous period’s per-capita land-use change persists into the next. Within countries, higher urban density, net in-migration, and higher GDP per capita all predict *less* per-capita land consumption — i.e., richer countries densify, not sprawl, when we look at the within-country margin that cross-sectional studies miss.

The remainder of the paper is structured as follows. Section 2 formally defines the SDG 11.3.1 indicator and its limitations. Section 3 introduces BpCR and demonstrates its properties. Section 4 describes the data. Section 5 outlines the empirical strategy. Section 6 presents the results, and Section 7 concludes.

2. The SDG 11.3.1 indicator and its limitations

$$\text{LCRPGR} = \frac{\text{LCR}}{\text{PGR}}, \quad \text{LCR} = \frac{V_t - V_{t-z}}{V_{t-z}} \cdot \frac{1}{z}, \quad \text{PGR} = \frac{\ln(P_t/P_{t-z})}{z}.$$

The arithmetic LCR over a logarithmic PGR makes the ratio asymmetric, and PGR in the denominator makes it diverge when population is roughly stable (Nicolau et al., 2019).

3. A stable metric: the Built-up per Capita Rate (BpCR)

$$\text{BpCR}_{i,t} = \frac{1}{z} \ln \left(\frac{V_t/P_t}{V_{t-z}/P_{t-z}} \right) = \text{LCR}_{\log} - \text{PGR}_{\log}.$$

4. Data

See Pesaresi et al. (2024), Schiavina et al. (2023), UN SNAAMA (2025) and UN DESA WPP (2024) for the source datasets.

Table 1: Estimation sample

Quantity	Value
Countries	193
Country-period observations	2116

5. Empirical strategy

$$\text{Metric}_{i,t} = \rho \text{Metric}_{i,t-1} + \beta_1 \ln(\text{UrbDens})_{i,t} + \beta_2 \text{UrbShare}_{i,t} + \beta_3 \ln(\text{GDPpc})_{i,t} + \beta_4 \text{NetMigr}_{i,t} + \mu_i + \delta_t + \varepsilon_{i,t}.$$

6. Results

7. Robustness

8. Discussion

9. Conclusion

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